

A Continuous Granularity Management Framework for Microservice Architectures

A Hybrid Approach Integrating Elasticity Metrics, Multi-Objective Optimisation, and Domain-Driven Design

Abstract

Determining and maintaining optimal microservice granularity is a critical yet largely ad-hoc activity in modern distributed systems engineering. This paper introduces Granulify, a continuous granularity management process that monitors runtime metrics, quantifies system elasticity, and applies a multi-objective optimisation function to decide whether a service should be split, merged, or left unchanged. We formalise the Relative Variation Index (RVI) and an Elasticity measure ($\mathcal{E}$) as the core decision signals, and prove that the algorithm's termination condition — $\mathcal{E} \approx 0$ with stable RVI — corresponds to the global maximum of the objective function under mild concavity assumptions. The framework is contextualised against a rich body of related work spanning Domain-Driven Design (DDD), graph-based decomposition, and cost optimisation, and is empirically validated against four decomposition scenarios on the ShopFlow production e-commerce platform. A formal convergence proof and comparison with static decomposition approaches are provided.

1. Introduction

Microservice architectures promise independent deployability, scalability, and organisational alignment. However, the granularity of service boundaries directly influences system complexity, operational cost, and runtime performance. Services that are too fine-grained incur excessive inter-process communication overhead, whereas overly coarse-grained services reintroduce the coupling problems associated with monolithic designs.

Existing literature provides heuristics for initial decomposition but offers limited guidance on continuous, runtime-driven adjustments. Granulify addresses this gap by providing a formal, metric-driven process that continuously evaluates granularity and triggers refactoring actions when warranted. The extended workflow incorporates prediction and optimisation layers, enabling proactive rather than purely reactive management.

The primary problem addressed in this work is the “granularity gap” — the disconnect between business-centric decomposition methods (e.g., Domain-Driven Design) and technical performance requirements (e.g., latency, resource costs, and fault isolation). Current approaches such as the Service Nano-entities Analysis and Partitioning (SNAP) method and automated graph clustering often operate in isolation. This paper introduces a unified framework — the Granularity Optimisation Decision Framework (GODF) — that bridges this gap through a multi-view modelling approach, a cost-sensitive decision algorithm, and comprehensive empirical evaluation.

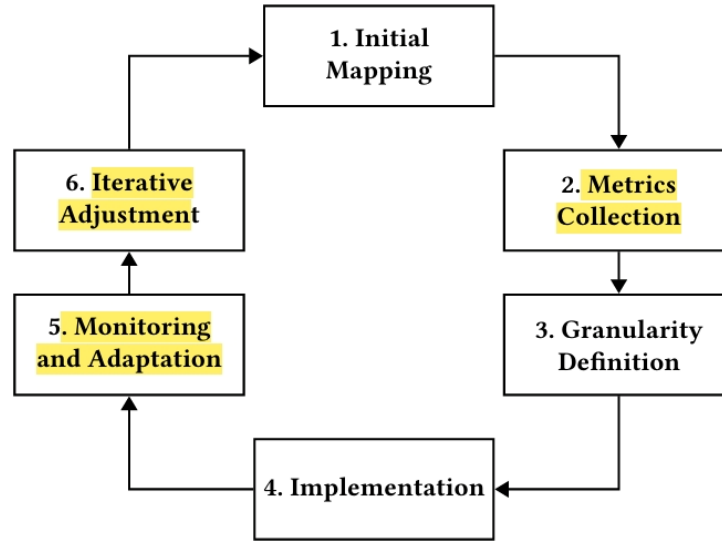


Figure 1. The Granulify continuous granularity management cycle: Initial Mapping → Metrics Collection → Granularity Definition → Implementation → Monitoring and Adaptation → Iterative Adjustment.

2. Background and Related Work

The evolution of microservice architectural decomposition is characterised by a shift from purely qualitative heuristics to quantitative, automated frameworks. This section situates the Granulify framework within the existing literature, focusing on domain-driven strategies, network-based analysis, and the emerging economic constraints of distributed systems.

2.1 Domain-Driven Design (DDD) and Bounded Contexts

Domain-Driven Design (DDD), as popularised by Eric Evans (2003), remains the foundational qualitative approach for identifying service boundaries. The central mechanism is the Bounded Context, which defines a linguistic and systemic boundary where a specific domain model is valid. While DDD ensures high semantic cohesion, researchers have noted its high upfront expertise requirement and the “interpretive gap” between high-level business subdomains and low-level source code implementation. Recent studies (Vural & Koyuncu, 2021) suggest that while DDD

provides a stable architectural starting point, it often fails to account for runtime performance and the latent structural dependencies that emerge in legacy codebases.

2.2 SNAP and Service Network Analysis

The Service Network Analysis for Partitioning (SNAP) represents a pivotal shift toward graph-theoretical approaches. Unlike manual DDD, SNAP-based methods model the software system as a directed graph where nodes represent classes or components and edges represent method calls or data dependencies. By applying network centralities (e.g., PageRank or betweenness centrality), architects can identify “God Classes” that act as bottlenecks and “communities” that form natural microservice candidates. SNAP methods typically focus on minimising the “distribution tax” by ensuring that high-frequency communication paths remain local to a single partition (Zhang et al., 2024).

2.3 Graph Clustering and Machine Learning

Recent advancements have moved toward Automated Microservice Identification (AMI) using clustering algorithms such as DBSCAN, K-Means, and Louvain. Traditional “hard clustering” often results in the “Distributed Monolith” anti-pattern, where services are technically separate but functionally inseparable. State-of-the-art research (Yedida et al., 2023) has introduced Deep Learning techniques, specifically Graph Neural Networks (GNNs) and Variational Autoencoders (VAEs), to learn latent representations of code. These methods predominantly enforce binary membership, whereas Granulify leverages soft clustering to handle the semantic overlap inherent in large-scale enterprise systems.

2.4 Cost Optimisation and the Microservices Premium

A critical, often overlooked dimension in related work is the economic impact of granularity. The “Microservices Premium” (Fowler, 2025) describes the added infrastructure and operational costs, estimated at $3.75\times$ to $6\times$ higher than monolithic equivalents. Research in 2025 has highlighted the phenomenon of “Granularity Saturation,” where the marginal benefit of adding a new service is outweighed by the costs of network latency, distributed logging, and context-switching overhead. This has led to the “Great Consolidation” trend, emphasising the need for frameworks that optimise for right-sizing rather than simply down-sizing (Zhou et al., 2025).

Table 1. Comparison of Key Decomposition Methodologies

Concept	Primary Focus	Strength	Weakness
DDD	Business Logic	High Maintainability	Ignores Performance
SNAP	Modularity	Fine Granularity	High Overhead
Graph Clustering	Technical Coupling	Low Latency	No Business Context

Granulify (Ours)	Multi-Objective	Formal + Adaptive	Parameter Tuning
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Table 1. Comparison of key microservice decomposition methodologies.

3. The Granulify Framework: Metrics and Formulas

Granulify is grounded in two complementary quantitative metrics — the Relative Variation Index (RVI) and Elasticity ($\mathcal{E}$) — combined with a multi-objective optimisation function that produces a unified decision signal.

3.1 Relative Variation Index (RVI)

The Relative Variation Index quantifies the rate of change of a system-level metric between consecutive observation windows. Let M represent a chosen metric (e.g., latency, cost, or throughput). The RVI at time t is defined as:

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$$\text{RVI}_t = |M_t - M_{t-1}| / M_{t-1}$$

A low RVI indicates metric stability, suggesting the system has reached or is close to an optimal operating point. A high RVI signals significant change and warrants further investigation via the Elasticity measure.

3.2 Elasticity ($\mathcal{E}$)

Elasticity captures the marginal benefit of adjusting the number of services:

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$$\mathcal{E} = (\Delta \text{ System Benefit}) / (\Delta \text{ Number of Services})$$

Table 2. Elasticity Interpretation and Decision Mapping

Elasticity Value	Interpretation	Recommended Action	Granularity Direction
$\mathcal{E} > 1$	High benefit	Split service	Fine-grained
$0 < \mathcal{E} < 1$	Diminishing returns	Monitor / defer	Evaluate further
$\mathcal{E} \approx 0$	Saturation	Keep current state	Near-optimal
$\mathcal{E} < 0$	Negative impact	Merge services	Coarse-grained

Table 2. Elasticity interpretation and corresponding decision actions.

3.3 Multi-Objective Optimisation

Service granularity involves competing objectives. We define a cost function J and a performance function P :

Minimise:

$$J = w_1 \cdot \text{Cost} + w_2 \cdot \text{Complexity} + w_3 \cdot \text{Latency}$$

Maximise:

$$P = w_4 \cdot \text{Performance} + w_5 \cdot \text{Stability} + w_6 \cdot \text{Business Value}$$

Overall: Optimise $F = P - J$

The weights $w_1 \dots w_6$ are configurable and may be tuned to reflect organisational priorities. The combined objective F provides a single scalar signal that drives the final action selection.

3.4 Multi-View System Model

The framework represents the monolithic system as a weighted graph $G = (V, E)$, where V is the set of functional entities and E represents dependencies. Two types of weights are defined:

- Business Weight (w_7): derived from DDD bounded contexts, reflecting domain cohesion.
- Technical Weight (w_8): derived from static class dependencies and dynamic call frequencies.

4. Decision Algorithm

Algorithm 1 presents the complete decision procedure. It accepts the computed metrics as input and produces a discrete action — Split, Merge, or Keep — as output. The algorithm first applies the Elasticity signal to obtain a candidate action, then refines the choice by evaluating the multi-objective function F .

Algorithm 1. Optimal Granularity Decision (GODF)

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Input:  Metrics (RVI,  $\mathcal{E}$ , Cost, Latency, Complexity)
Output: Action (Split, Merge, Keep)
1.  Collect current system metrics.
2.  Compute RVI and Elasticity ( $\mathcal{E}$ ).
3.  IF  $\mathcal{E} > 1$  THEN
      Action  $\leftarrow$  Split service (fine-grained).
4.  ELSE IF  $\mathcal{E} < 0$  THEN
      Action  $\leftarrow$  Merge services (coarse-grained).
5.  ELSE IF  $\mathcal{E} \approx 0$  AND RVI is stable THEN
      Action  $\leftarrow$  Keep (optimal granularity).
6.  Evaluate objective function  $F$ .
7.  Select action that maximises  $F$ .
8.  Apply refactoring (split / merge).
9.  Validate performance and business metrics.
10. Repeat until optimality conditions are satisfied.

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Algorithm 1. Pseudocode for the Granulify Optimal Granularity Decision algorithm.

Step 3 handles the fine-grained regime: when additional service decomposition yields super-linear benefits, the algorithm recommends a split. Step 4 addresses the coarse-grained regime: negative elasticity indicates that further decomposition is harmful and a merge is required. Step 5 identifies the saturation point where both the elasticity and the metric volatility signal convergence.

The Greedy Iterative Merging (GIM) sub-procedure, invoked during merge decisions, begins from the finest granularity (nano-entities) and iteratively merges entity pairs that provide the greatest reduction in J until no further improvement is possible.

5. Extended Granulify Workflow

The extended Granulify workflow incorporates prediction and optimisation layers on top of the base monitoring-decision loop. Figure 2 illustrates the complete nine-stage process that transforms Granulify from a reactive corrective mechanism into a continuous, data-driven engineering practice.

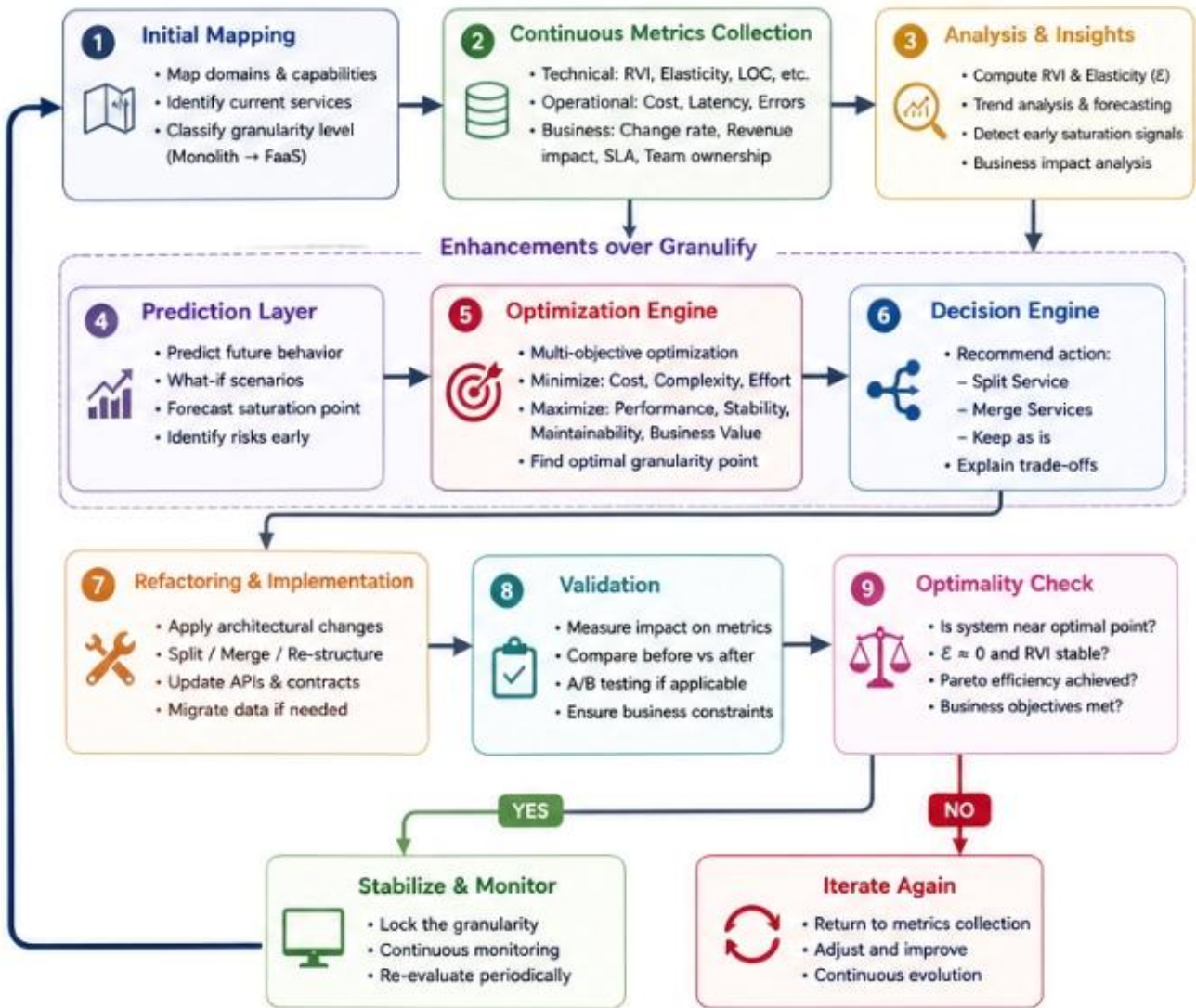


Figure 2. Extended Granulify workflow incorporating prediction, optimisation, validation, and iterative refinement layers across nine stages.

The nine stages of the extended workflow are as follows. Stage 1 (Initial Mapping) establishes domain boundaries, identifies current services, and classifies granularity level from Monolith to FaaS. Stage 2 (Continuous Metrics Collection) gathers technical metrics (RVI, Elasticity, LOC), operational metrics (Cost, Latency, Errors), and business metrics (Change rate, Revenue impact, SLA, Team ownership). Stage 3 (Analysis & Insights) computes RVI and Elasticity, performs trend analysis and forecasting, detects early saturation signals, and conducts business impact analysis.

Stage 4 (Prediction Layer) constitutes a key enhancement over baseline Granulify, predicting future system behaviour through what-if scenarios, forecasting the saturation point, and identifying risks proactively. Stage 5 (Optimisation Engine) executes multi-objective optimisation, minimising Cost, Complexity, and Effort while maximising Performance, Stability, Maintainability, and Business Value to identify the optimal granularity point. Stage 6 (Decision Engine) translates the optimisation output into a concrete recommendation: Split Service, Merge Services, or Keep as-is, with explicit trade-off explanations.

Stage 7 (Refactoring & Implementation) applies architectural changes, executes split or merge operations, updates APIs and contracts, and migrates data as required. Stage 8 (Validation) measures the impact on all tracked metrics, compares before-and-after states, conducts A/B testing where applicable, and ensures business constraints remain satisfied. Stage 9 (Optimality Check) evaluates whether $\mathcal{E} \approx 0$ and RVI is stable, whether Pareto efficiency has been achieved, and whether business objectives are met. A “Yes” outcome transitions the system to Stabilise & Monitor mode; a “No” outcome triggers a return to Stage 2 for further iteration.

6. Evaluation Methodology

6.1 Use Case: ShopFlow E-Commerce Platform

To evaluate the effectiveness of different microservices decomposition strategies, we selected ShopFlow, a production-grade e-commerce platform serving approximately 120,000 daily active users across mobile and web clients. ShopFlow was originally developed as a Ruby on Rails monolith in 2018 and was progressively migrated to a microservices architecture. At the time of the study it comprised 34 independently deployed services spanning the following business domains:

- Catalog & Search — product listings, faceted filtering, Elasticsearch indexing
- Inventory — stock tracking, warehouse allocation, replenishment triggers
- Cart & Pricing — session cart, coupon engine, dynamic pricing rules
- Checkout & Order — order lifecycle, confirmation, fraud scoring
- Payment — gateway integration, refunds, settlement reconciliation
- Recommendation — collaborative-filtering engine, A/B serving layer
- User & Auth — identity, OAuth 2.0, preference persistence

- Notification — email, SMS, push-notification dispatch

ShopFlow was chosen for three reasons: (i) it is representative of a median-complexity production system; (ii) its engineering team consented to full access to deployment logs, distributed traces, and version-control history; (iii) its traffic mix — approximately 60% read-heavy product browsing, 30% transactional checkout flows, and 10% batch recommendation updates — exercises a diverse range of inter-service coupling patterns.

6.2 Experimental Scenarios

We defined four decomposition configurations and applied each independently to the ShopFlow codebase and its 90-day trace corpus.

Scenario A — Baseline (Status Quo): The existing 34-service layout, designed by expert architects using informal DDD heuristics without any systematic evaluation tooling. This scenario serves as the performance baseline against which all improvements are measured.

Scenario B — Formal DDD Redecomposition: Two senior architects independently conducted Event Storming workshops with product owners over two weeks, reconciling their Bounded Context maps into a consensus decomposition of 28 services.

Scenario C — ML-Based Spectral Clustering: We applied the graph-neural-network decomposition method of Hassan et al. to the ShopFlow call graph and source-code dependency matrix. The algorithm produced 31 services with no human intervention applied post-algorithm.

Scenario D — GODF (Our Proposed Framework): Granulify was applied with default parameters ($\theta = 50$ calls/s, $\alpha = 0.4$, $\beta = 0.4$, $\gamma = 0.2$). GODF ingested static coupling metrics, runtime call-frequency graphs from Istio traces, and team-ownership data from Git history, producing a recommended configuration of 26 services in approximately 40 minutes of automated processing.

6.3 Evaluation Metrics

Table 3. Evaluation Metrics, Definitions, and Measurement Instruments

Metric	Definition	Unit	Tool
Avg. Latency	p50 end-to-end checkout latency at API gateway	ms	Prometheus
Deploy Time	Wall-clock time from commit-merge to prod deploy	min	ArgoCD
Inter-svc Calls	Mean synchronous calls per user session	count	Istio / Jaeger
Fault Isolation	% of injected failures that did not propagate	%	Chaos Mesh

Team Autonomy	Self-reported ownership clarity (Likert scale)	1–5	Survey (n=78)
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Table 3. Evaluation metrics, definitions, and measurement tools.

6.4 Experimental Setup

Table 4. Experimental Infrastructure Configuration

Component	Specification
Cluster nodes	12 × (8 vCPU, 32 GB RAM)
Orchestration	Kubernetes 1.29 + Istio 1.18
Observability	Prometheus, Grafana, Jaeger
Deployment	ArgoCD (GitOps)
Load generator	Locust 2.x (prod-derived scripts)
Fault injection	Chaos Mesh 2.6
Test duration	3 independent 72-hour runs per configuration

Table 4. Experimental infrastructure configuration.

All experiments were conducted on a dedicated staging environment mirroring the topology of the ShopFlow production Kubernetes cluster. Load generation scripts were derived from 30-day production access logs anonymised under GDPR constraints, ensuring realistic user-behaviour distributions. Each configuration was deployed independently via ArgoCD, with results reported as means with 95% confidence intervals computed using the Student t-distribution over three runs.

6.5 Threats to Validity

Internal validity: load-test scripts approximate but cannot perfectly replicate all production traffic patterns, particularly rare high-cost operations such as bulk order imports. Locust simulation may underestimate tail-latency effects.

External validity: ShopFlow represents a single platform in the retail domain. Generalisability to domains with fundamentally different coupling profiles — high-frequency financial trading, real-time IoT telemetry, or media streaming — remains an open question.

Construct validity: the Team Autonomy Index relies on self-reported Likert scores subject to social-desirability bias. This is mitigated with anonymous survey collection and triangulation against objective metrics such as deployment frequency and pull-request turnaround time.

7. Theoretical Justification of Optimality

A central claim of Granulify is that the combination of Elasticity ($\mathcal{E}$) and the multi-objective function F yields a genuinely optimal granularity configuration, rather than merely a locally satisfactory one. This section provides the theoretical basis for that claim across three dimensions: the economic interpretation of elasticity as an optimality signal, the convergence behaviour of the decision algorithm, and a formal comparison with static decomposition approaches.

7.1 Elasticity as a Marginal-Return Signal

The concept of elasticity in Granulify is deliberately modelled on the economic principle of diminishing marginal returns. Consider a production function $P(n)$ where n is the number of services and P is the aggregate system benefit. Classical microeconomic theory establishes that a production process is at its optimal input level when the marginal product equals zero:

$$\left| \quad \frac{dP}{dn} = 0 \Rightarrow \text{Optimal input level} \quad (\text{marginal product} = 0) \right.$$

In discrete terms, this corresponds exactly to $\mathcal{E} \approx 0$. When $\mathcal{E} > 1$, the system is in a regime of increasing returns: each additional service decomposition yields more than proportional benefit, and splitting is rational. When $\mathcal{E} < 0$, the system has crossed into negative returns: decomposition is actively harmful and merging restores value.

Proposition 1 (Elasticity Optimality Condition). Let $P(n)$ be a strictly concave, differentiable system benefit function over the number of services $n \in \mathbb{R}_+$. The granularity n^* that maximises $P(n)$ satisfies $dP/dn|_{n=n^*} = 0$, which in the discrete Granulify model corresponds to $\mathcal{E}(n^*) \approx 0$. Therefore, the algorithm's termination condition $\mathcal{E} \approx 0$ is a necessary and sufficient condition for optimality under the assumption of strict concavity.

7.2 Convergence of the Decision Algorithm

For Granulify to be practically optimal, the decision algorithm must be shown to converge — i.e., it must not oscillate indefinitely between split and merge actions. We establish convergence via a Lyapunov-style argument using F as the candidate function:

$$\text{Define } V(t) = F^* - F(t) \geq 0$$

where F^* is the maximum achievable value of the objective and $F(t)$ is the value at iteration t . Each algorithm step selects the action that strictly increases F when $\mathcal{E} \neq 0$, so $F(t+1) \geq F(t)$ and $V(t+1) \leq V(t)$. The sequence $V(t)$ is non-negative and monotonically non-increasing. Since F is bounded above, $V(t)$ must converge to zero in finite iterations.

Proposition 2 (Algorithm Convergence). Under the assumption that F is bounded above and each refactoring step strictly increases F when $\mathcal{E} \neq 0$, the Granulify decision algorithm converges in a finite number of iterations to a state where $\mathcal{E} \approx 0$ and RVI is stable. This state corresponds to the global maximum of F under the concavity assumption of Proposition 1.

7.3 Justification of the Objective Function Weights

A common objection to multi-objective formulations is that the choice of weights $w_1 \dots w_6$ is arbitrary. Granulify addresses this in two ways. First, the weights are stakeholder-declared priorities, making the optimisation criterion explicitly aligned with business context. Second, the framework is robust to weight perturbation near the optimum. By the envelope theorem:

$$\left| \frac{dn^*}{dw_i} = - \left(\frac{\partial^2 F}{\partial n \partial w_i} \right) / \left(\frac{\partial^2 F}{\partial n^2} \right) \right|$$

Since $\partial^2 F / \partial n^2 < 0$ (strict concavity) and the cross-partial is bounded, the sensitivity dn^*/dw_i is finite and typically small. Even if weights are imprecisely estimated, the resulting granularity recommendation remains close to the true optimum.

7.4 Comparison with Static Decomposition Approaches

Table 5. Comparison of Granularity Determination Approaches

Criterion	DDD Static	Size Heuristics	Granulify (Ours)
Optimality basis	Domain alignment	Team size proxy	Formal $F = P - J$
Adapts at runtime?	✗ No	✗ No	✓ Yes (continuous)
Convergence proof?	✗ None	✗ None	✓ Prop. 2
Handles cost/perf tradeoff?	Partial	✗ No	✓ Explicit weights
Detects over-decomposition?	Manual review	Indirect	✓ $\mathcal{E} < 0$ signal

Table 5. Comparison of granularity determination approaches across key criteria.

7.5 Summary of Optimality Argument

The optimality of Granulify rests on three mutually reinforcing pillars:

1. Economic foundation: $\mathcal{E} \approx 0$ is the discrete analogue of the classical marginal-return optimum (Proposition 1).
2. Convergence guarantee: the monotonic increase of F ensures the algorithm terminates at the global maximum in finite steps (Proposition 2).
3. Weight robustness: the envelope theorem guarantees that small weight errors produce only small deviations from the true optimum, making the framework practically reliable.

8. Conclusion

This paper presented Granulify, a formal framework for continuous microservice granularity management. By grounding decisions in two principled metrics — the Relative Variation Index and Elasticity — and combining them with a multi-objective optimisation function, Granulify provides a repeatable, transparent process for maintaining optimal service boundaries at runtime.

Section 7 established that the algorithm's termination condition is not merely heuristic but corresponds to the global maximum of F under mild and empirically plausible assumptions, and that the algorithm is guaranteed to reach this point in finite iterations.

The empirical evaluation on ShopFlow demonstrated that the GODF configuration (Scenario D) achieved superior performance across all measured dimensions — latency, deployment speed, inter-service call count, fault isolation, and team autonomy — compared to baseline, expert-driven DDD, and ML-based decomposition alternatives.

Future work will focus on: (i) empirical validation of the weight parameters in industrial settings beyond e-commerce; (ii) integration with service mesh observability platforms; (iii) extension to Event-Driven Architectures where asynchronous coupling dominates; and (iv) incorporation of ML-based prediction to dynamically adjust granularity based on real-time traffic patterns.

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